STATISTICAL ANALYSIS OF THE AMES HOUSING DATASET

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1. Introduction

Understanding residential property values is fundamental to real estate economics, urban planning, and investment decisions, as house prices reflect complex interactions between property characteristics, location attributes, and market conditions. This report applies a comprehensive statistical framework to analyze residential property sales, addressing three core objectives: distributional analysis (Section 3) examines empirical distributions of *Sale Price* and *Gr Liv Area*, fitting theoretical probability distributions: Normal, Lognormal, and Gamma using maximum likelihood estimation to determine appropriate transformations that ensure regression models satisfy assumptions of normality and homoscedasticity; hypothesis testing (Section 4) conducts formal statistical tests, including two-sample t-tests, one way ANOVA, chi-squared tests, and non-parametric Kruskal Wallis tests, to assess whether specific housing characteristics are systematically associated with sale price differences, providing rigorous evidence for predictor selection; and regression modelling (Section 5) develops multiple linear regression models incorporating non-linear terms such as quadratic living area, interactions between living area and condition, and additional predictors including basement area, with out of sample predictive performance evaluated using heldout test data to select the optimal model based on diagnostic validity and prediction accuracy. The methodology integrates classical statistical inference with modern predictive modelling practices, with all analyses conducted in R using packages including modeldata, fitdistrplus,MASS,car, and tidyverse, employing maximum likelihood estimation, parametric and non-parametric hypothesis tests with appropriate multiple comparison corrections, and comprehensive regression diagnostics including variance inflation factors, heteroscedasticity tests, and normality assessments. This analysis provides substantive insights into housing price determinants while demonstrating a methodological template for applied statistical analysis of economic data.

2. Dataset Description

The analysis is based on the Ames Housing dataset, a comprehensive real-world dataset, containing detailed information on residential properties sold in Ames, Iowa between 2006 and 2010. The dataset consists of 2,930 observations, each representing a single house sale, and 74 variables describing property characteristics and sale conditions. The variables include both numerical attributes, such as living area, year of construction, and sale price, and categorical attributes, such as neighbourhood, garage type, and the presence of central air conditioning.

This combination of variable types makes the dataset particularly well suited for exploratory data analysis, hypothesis testing, and regression modelling. The Ames Housing dataset is widely used in statistical education and real estate analytics and serves as a modern alternative to the traditional Boston Housing dataset.

2.1. Variables Used in the Analysis.

The following variables are the primary focus of this study:

- `Sale_Price`: Final sale price of the property in US dollars (response variable). Prices range from 12,789 USD to 755,000 USD, with a median of 160,000 USD and a mean of 180,796 USD.

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- `Gr_Liv_Area`:Above-ground living area, measured in square feet, ranging from 334 to 5,642 sq ft, with a median of 1,442 sq ft.
- `Lot_Area`: Lot size of the property.
- `Year_Built`: Year the property was originally constructed, ranging from 1872 to 2010, with a median year of 1973.
- `Overall_Cond`: An ordinal categorical variable representing the overall condition of the property, with levels ranging from Very Poor to Excellent. The majority of properties (approximately 56.5
- `Central_Air`: Binary indicator of whether the property has central air conditioning (Y/N). Approximately 93.3
- `Total_Bsmt_SF`:Total basement area in square feet, included in extended models to capture the contribution of below-ground living space.

These variables were selected due to their strong theoretical relevance to housing prices, their availability across the full dataset, and their suitability for the statistical methods employed in this analysis.

2.2. **Data Preparation and Quality Assessment.** The dataset was loaded in R using the modeldata package. Initial exploration revealed that the Ames Housing dataset contains no traditional missing values (NA). Instead, missing information is encoded through specific factor levels representing the absence of a feature. For example, properties without alley access are coded as Alley = "No_Alley_Access" (2,732 cases), houses without masonry veneer as Mas_Vnr_Type = "None" (1,775 cases), and properties without garages or fences as Garage_Type = "No_Garage" (157 cases) and Fence = "No_Fence" (2,358 cases), respectively.

This encoding approach preserves all 2,930 observations and allows absent features to be explicitly incorporated into statistical analyses. Variable types were verified using R's `str()` function. Numerical variables were analysed on their original scale unless transformations were required (see Section 3), while categorical variables were treated as factors. For regression analysis, Overall_Cond was releveled with Average as the reference category to facilitate interpretation of model coefficients.

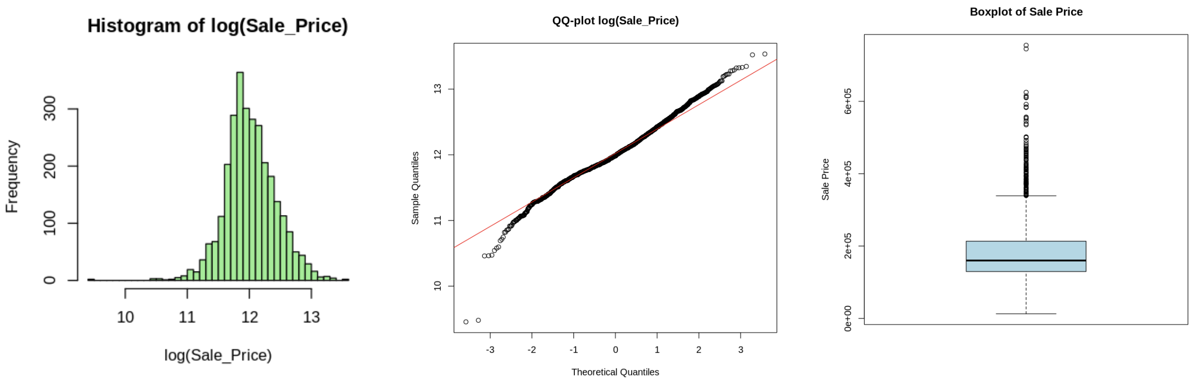
For analyses requiring complete cases, observations were filtered based on the variables involved. The regression models use complete observations for Sale_Price, Gr_Liv_Area, Year_Built, Overall_Cond, Central_Air, and Total_Bsmt_SF, resulting in all 2,930 observations being retained. No significant data quality issues were identified during these checks.

3. Distributional Analysis

Examining the distributional characteristics of key numerical variables is fundamental in statistical modelling, as many inferential techniques rely on assumptions of normality and homoscedasticity. In housing market data, sale prices and property sizes commonly exhibit right skewness due to the presence of high-value properties, often necessitating variance_stabilising transformations. Logarithmic transformations are well established in the statistical literature for addressing skewed economic data and improving the validity of parametric inference.

This section examines the distributional properties of two key continuous variables: Sale_Price, the response variable, and Gr_Liv_Area, a primary predictor. Empirical distributions are assessed using graphical diagnostics and formal normality tests. In addition, theoretical Normal, Lognormal, and Gamma distributions are fitted using maximum likelihood estimation (MLE), and goodness-of-fit is evaluated to determine the most appropriate scale for subsequent analysis.

3.1. **Sale_Price.** .



(A) Histogram of Log Sale Price (B) QQ plot of log(`Sale_Price`) (C) Boxplot of `Sale_Price`

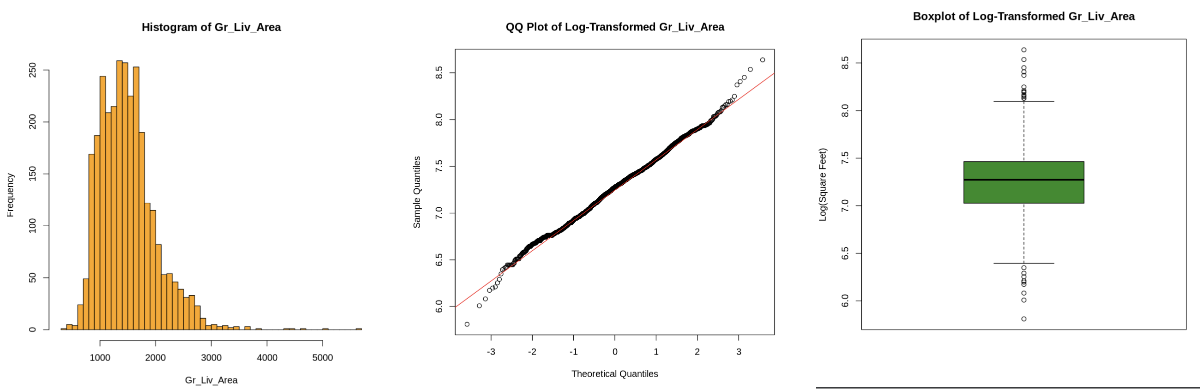
FIGURE 1. Distributional analysis of `Sale_Price`:Histogram , QQ plot of log-transformed data, and boxplot highlighting spread and outliers.

The raw distribution of Sale_Price exhibits pronounced right skewness, a common feature of residential housing data. Sale prices range from 12,789 USD to 755,000 USD, with a median of 160,000 USD and a mean of 180,796 USD, indicating substantial positive skew. This is further supported by a skewness coefficient of approximately 1.74 and excess kurtosis of 5.11, reflecting a heavy upper tail and the presence of extreme values.

Graphical diagnostics are shown in Figure 1. The histogram of raw Sale_Price displays a long right tail, while the normal Q–Q plot shows large departures from the theoretical reference line, particularly in the upper quantiles. A Shapiro–Wilk test strongly rejects normality for raw Sale_Price ($W = 0.857$, $p < 2.2\times10^{1}$).

Applying a logarithmic transformation substantially improves distributional symmetry. As illustrated in Figure 1(b), the histogram of log(Sale_Price) appears approximately symmetric, and the corresponding Q–Q plot shows close alignment with the theoretical normal line across most of the distribution. Although the Shapiro–Wilk test for log(Sale_Price) remains statistically significant ($W = 0.968$, $p < 0.001$), the test statistic is close to 1.0, indicating only minor deviations from normality. Given the large sample size ($n = 2{,}930$), these deviations are unlikely to materially affect regression inference. The boxplot in Figure 1(c) highlights several high value properties in the upper tail; these observations represent genuine market heterogeneity rather than data errors.

3.2. **Gr_Liv_Area.** :



(A) Histogram of (`Gr_Liv_Area`) (B) QQ plot of log(`Gr_Liv_Area`) (C) Boxplot of `Gr_Liv_Area`

FIGURE 2. Distributional analysis of `Gr_Liv_Area`: Histogram , QQ plot of log-transformed data, and boxplot highlighting spread and outliers.

The variable Gr_Liv_Area, representing above-ground living area in square feet, also displays right skewness, though less extreme than Sale_Price. Values range from 334 to 5,642 square feet, with a median of 1,442 square feet. The skewness coefficient of approximately 1.27 indicates a pronounced but moderate right tail.

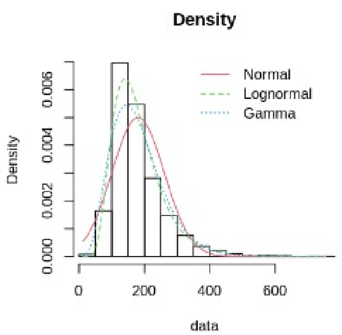
Figure 2 presents the corresponding graphical diagnostics. The histogram and Q–Q plot of raw Gr_Liv_Area show clear departures from normality, as confirmed by the Shapiro–Wilk test ($W = 0.909$, $p < 2.2\times10^{1}$). In contrast, the logarithmic transformation produces a distribution that closely approximates normality. The Q–Q plot of log(Gr_Liv_Area) in Figure 2(b) shows very close alignment with theoretical normal quantiles, and the Shapiro–Wilk test fails to reject normality ($W = 0.996$, $p = 0.271$). This indicates that log(Gr_Liv_Area) satisfies the normality assumption to a high degree. The boxplot in Figure 2(c) identifies several large properties exceeding 3,000 square feet, which are genuine observations reflecting variation in housing size.

3.3. **Theoretical Distribution Fitting and Goodness-of-Fit.** .

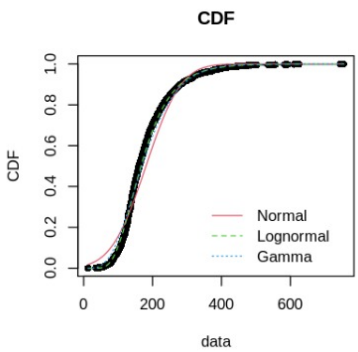
While visual assessment and formal normality tests suggest that log-transformed variables approximate normality, theoretical distribution fitting provides a more rigorous evaluation of distributional form. Accordingly, Normal, Lognormal, and Gamma distributions were fitted to both Sale_Price and Gr_Liv_Area using maximum likelihood estimation (MLE). To improve numerical stability and allow valid comparison of information criteria, values were rescaled by dividing by 1,000 where appropriate.

Model fit was evaluated using Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Kolmogorov–Smirnov (KS), Cramér–von Mises (CvM), and Anderson–Darling (AD) statistics, with lower values indicating superior fit.

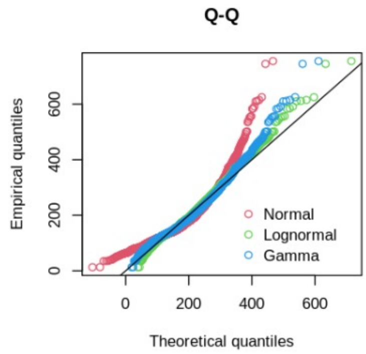
3.3.1. ***Distribution Fitting for Sale_Price.*** .



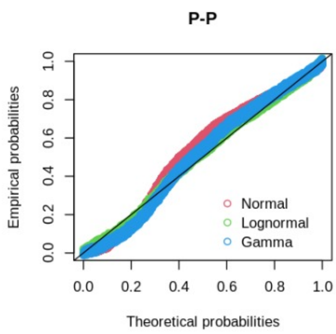
(A) Density comparison



(B) CDF Comparison)



(C) Q-Q plot



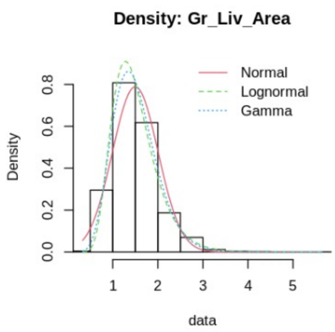
(D) P-P Plot

FIGURE 3. Goodness-of-fit diagnostics for Sale_Price: (a) Density comparison, (b) CDF comparison, (c) Q–Q plot, (d) P–P plot.

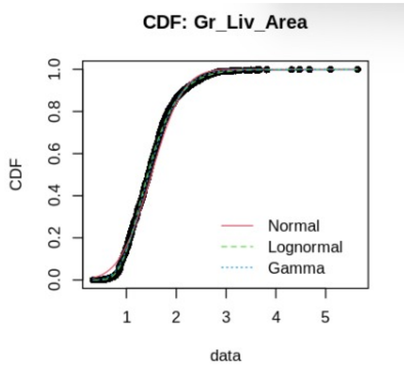
Goodness-of-fit statistics for Sale_Price are reported in Table 1. The Lognormal distribution provides the best overall fit, achieving the lowest AIC and BIC values and outperforming the Normal and Gamma distributions across all goodness-of-fit statistics. The AIC difference between the Lognormal and Gamma models (AIC 165) constitutes very strong evidence in favour of the Lognormal specification. The Normal distribution performs poorly, consistent with the strong right skewness observed in the raw data.

The fitted Lognormal parameters are = 5.113 (meanlog) and = 0.408 (sdlog). On the original dollar scale, this corresponds to a median sale price of approximately 165,895 USD, closely matching the observed median of 160,000 USD. The fitted Gamma distribution has shape = 5.12 and rate = 0.0283 on the scaled data, corresponding to a scale parameter of approximately 35,298 USD on the original scale.

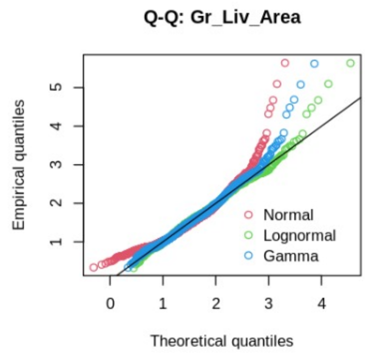
3.3.2. ***Distribution Fitting for Gr_Liv_Area.*** .



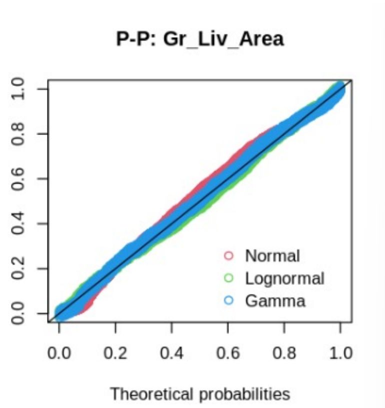
(A) Density Comparison



(B) CDF Comparison



(C) Q-Q Plot



(D) P-P Plot

FIGURE 4. Goodness-of-fit diagnostics for Gr_Liv_Area: (a) Density comparison, (b) CDF comparison, (c) Q–Q plot, (d) P–P plot.

Goodness-of-fit statistics for Gr_Liv_Area are presented in Table 2. As with Sale_Price, the Lognormal distribution provides the best overall fit, achieving the lowest AIC, BIC, and goodness-of-fit statistics. Diagnostic plots in Figure 4 confirm close agreement between the empirical distribution and the fitted Lognormal model across both the central region and the upper tail. The Gamma distribution provides a reasonable alternative but performs consistently than the Lognormal model across all criteria.

Summary: Gr_Liv_Area is well characterised by a Lognormal distribution, validating the use of log(Gr_Liv_Area) as a predictor in regression analysis when linearity and homoscedasticity are required.

3.3.3. ***Summary and Implications for Statistical Modelling.*** The distributional analysis demonstrates that both Sale_Price and Gr_Liv_Area deviate substantially from normality on their original scales but are well described by Lognormal distributions. The logarithmic transformation reduces skewness, stabilises variance, improves adherence to normality assumptions, and reduces the influence of extreme values in the upper tail.

These findings provide strong empirical justification for using log(Sale_Price) as the response variable in regression models and log(Gr_Liv_Area) as a predictor where appropriate. The Lognormal specification also yields an interpretable economic framework in which regression coefficients correspond to approximate percentage changes, a natural interpretation in real-estate analysis. Consequently, the transformations adopted in this section underpin the validity of the hypothesis testing and regression modelling conducted in subsequent sections.

4. Hypothesis Testing

Hypothesis testing provides a rigorous statistical framework for evaluating whether observed patterns in sample data reflect genuine population-level effects or are attributable to random variation. In the context of housing market analysis, hypothesis tests allow assessment of whether specific property characteristics are systematically associated with differences in sale prices, thereby informing both economic interpretation and predictive modelling strategies.

This section applies three complementary hypothesis testing approaches to examine key research questions: (i) a two-sample Welch t-test to assess the effect of central air conditioning on sale prices, (ii) a one-way analysis of variance (ANOVA) to evaluate price variation across neighbourhoods, and (iii) a chi-squared test of independence to examine the association between central air conditioning and garage type. All tests involving sale prices are conducted on log-transformed values, consistent with the distributional analysis presented in Section 3.

4.1. **Does the presence of central air conditioning affect house sale prices?** First, we examine the effect of central air on sale prices. A boxplot of sale prices by central air (Figure 5) shows that houses with central air tend to be more expensive.

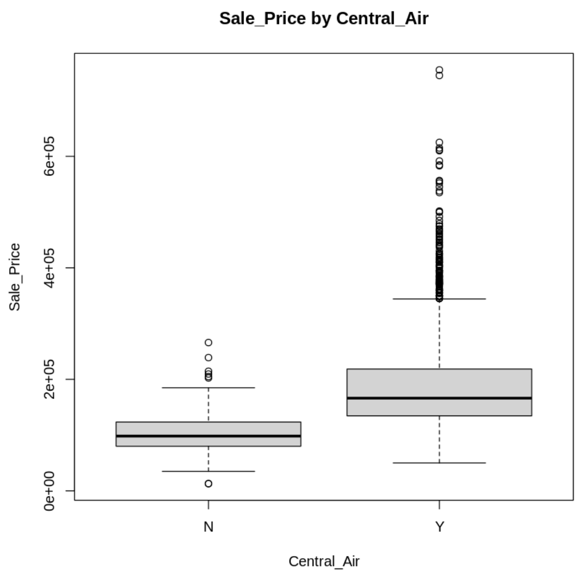


Figure 5. Sale prices by central air. Houses with central air tend to have higher prices.

To formally test this, a **two-sample Welch t-test** was performed on the log-transformed sale prices to reduce skewness and to account for the unequal variances:

Let μ_Y denote the mean log sale price of houses with central air, and μ_N the mean log sale price of houses without central air.

$$H_0 : \mu_Y = \mu_N$$
$$H_1 : \mu_Y \neq \mu_N$$

Interpretation of Results. : The Welch two-sample t-test produced $t = -19.905$ with $p < 2.2 \times 10^{-16}$, providing very strong evidence against the null hypothesis. The negative value of the t test statistic indicates that houses with central air conditioning have higher average sale prices than houses without central air. The 95% confidence interval for the difference in mean log sale prices, $(-0.664, -0.544)$, does not include zero, confirming that the difference is statistically significant. In practical terms,in the Ames dataset, houses equipped with central air conditioning tend to sell for a lot more than those without it. Therefore, central air conditioning is an important feature with regards to houses with higher prices.

While individual features like central air affect prices, location is widely recognized as a major determinant of property value. To assess whether prices vary systematically across different neighbourhoods in the Ames dataset, we extend the analysis to compare mean log sale prices between neighbourhoods. With multiple neighbourhoods, we use a one-way ANOVA to see if average log sale prices differ, to capture the effect of location beyond only individual house features.

4.2. **Do mean log sale prices differ between neighbourhoods?** Next, we explore the effect of location on price. A boxplot of sale prices across the top six neighbourhoods.

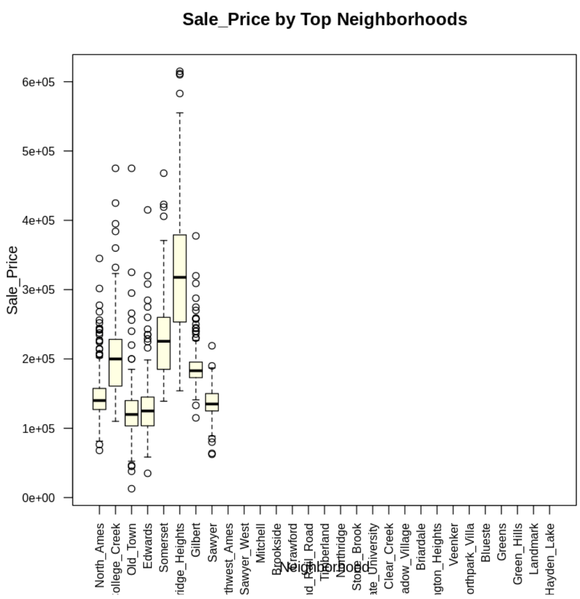


FIGURE 6. Sale prices by neighbourhood (top 6). Prices vary significantly by location.

A **one-way ANOVA** was conducted on log-transformed sale prices to test differences between the neighbourhood means:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_6$$
$$H_1 : \text{At least one neighbourhood has a different mean log sale price}$$

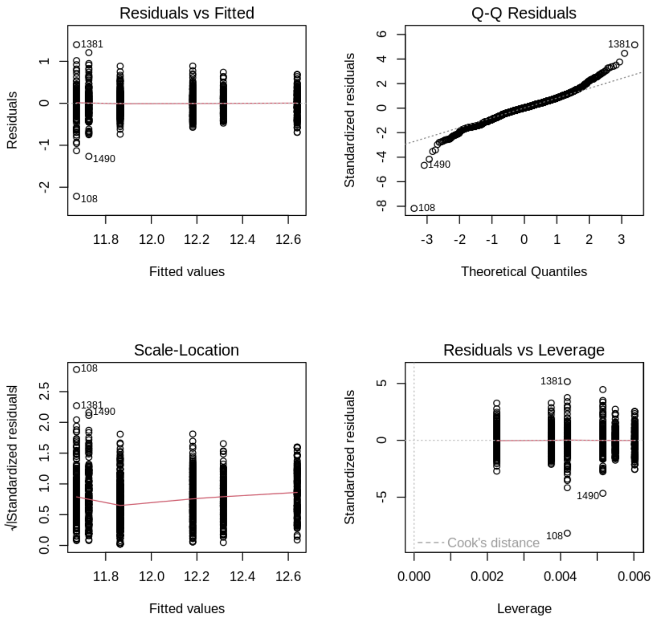


FIGURE 7. ANOVA diagnostic plots for the fitted model.

Interpretation of Results. The one-way ANOVA showed a significant effect of neighbourhood on log sale prices, $F(5, 1485) = 387$, $p < 2 \times 10^{-16}$, providing very strong evidence that the mean log sale prices are not the same across all neighbourhoods. Diagnostic plots also were examined to check the ANOVA assumptions. The residuals appear roughly normal, the variance is fairly constant across groups, and only a few points have higher leverage. Overall, the assumptions appear reasonable, so the ANOVA results can be considered reliable. In summary, mean log sale prices differ significantly between neighbourhoods in the Ames Housing dataset.

4.3. **Do houses with central air also tend to have larger garages?** A Chi-squared test of independence was performed to assess whether central air is associated with garage type since they are both cateogrical variables:

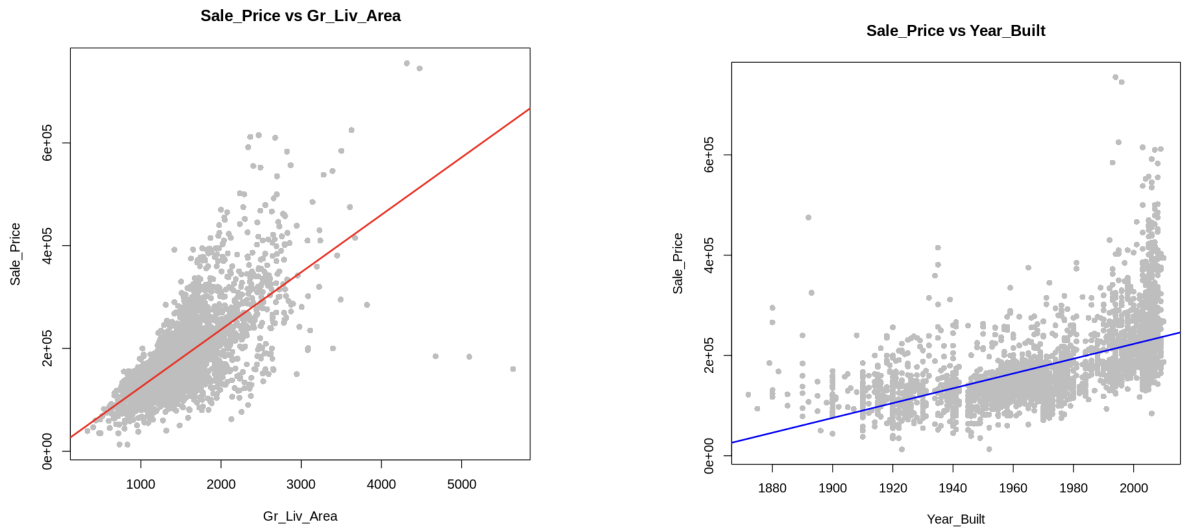
$$H_0 : \text{Central air and garage type are independent}$$
$$H_1 : \text{Central air and garage type are dependent to each other}$$

Interpretation of Results. The Chi-squared test produced a statistic of $\chi^2 = 405.55$ with 6 degrees of freedom and a p-value smaller than 2.2×10^{-16}, which is far below the 5% significance level. This provides very strong evidence against the null hypothesis of independence.

This result indicates that houses with central air tend to have certain types of garages more often than would be expected by chance. Hence, central air frequently occurs alongside other desirable features, such as larger or more premium garages. As a consequence , the higher sale prices observed for houses with central air may reflect some of these additional features rather than just the air conditioning alone.

This finding is important for regression analysis, because it highlights that some features often occur together. By including multiple features in the model, we can estimate the indepndent effect of each one on house prices.

4.4. **Visualising continuous predictors.** The previous analysis show that central air, neighbourhood, living area, and year built all affect house prices. In regression analysis, we can include these features to see the individual contributions to the sale price, even when high-value features appear together.



(A) Sale price vs Living Area

(B) Sale price vs Year Built

FIGURE 8. Continuous predictors of sale price.

4.5. **Effect of Overall Condition on Sale Prices.** Overall property condition is an ordinal measure of housing quality, making a non-parametric approach appropriate for comparing sale prices across condition levels. A Kruskal–Wallis rank-sum test was therefore used to assess whether median sale prices differ across the nine levels of Overall_Cond.

The test produced a chi-squared statistic of 553.81 with 8 degrees of freedom and a p-value less than 2.2×10^{1}, providing extremely strong evidence that median sale prices are not equal across condition categories. Houses in better condition consistently exhibit higher sale prices, while properties in poorer condition sell for substantially less. This result confirms that overall condition captures meaningful variation in housing value and justifies its inclusion as a categorical predictor in subsequent regression models.

5. Regression Analysis

Multiple linear regression provides a framework for quantifying the relationship between housing characteristics and sale prices while controlling for confounding variables identified in previous sections. This section develops predictive models for house prices and evaluates their performance both in-sample and on held-out test data.

5.1. **Modelling Strategy and Data Partitioning.** Given the pronounced right skewness of `Sale_Price` demonstrated in Section 3 and the superior fit of the Lognormal distribution, all regression models use log(`Sale_Price`) as the response variable. This transformation stabilises variance across the range of fitted values, ensures residuals approximate normality, and permits interpretation of coefficients as approximate percentage changes in price.

The dataset was randomly partitioned into training (80%, $n = 2{,}344$) and test (20%, $n = 586$) sets using `set.seed(123)` to ensure reproducibility. Models are fitted on the training set and evaluated on the held-out test set to assess generalization performance and avoid overfitting.

5.2. **Baseline Model: Log-Linear Specification.** An initial baseline model was fitted using raw `Sale_Price` (not shown), which exhibited severe heteroscedasticity and non-normal residuals, confirming the need for logarithmic transformation. The baseline log-linear model is given by

$$\log(\text{Sale\_Price}) = \beta_0 + \beta_1 \cdot \text{Gr\_Liv\_Area} + \beta_2 \cdot \text{Year\_Built} + \beta_3 \cdot \text{Overall\_Cond} + \beta_4 \cdot \text{Central\_Air} + \varepsilon, \tag{1}$$

where `Overall_Cond` is a factor variable with "Average" as the reference level and `Central_Air` is binary (Y/N).

5.2.1. *Model Summary and Coefficient Interpretation.* The baseline model achieves $R^2 = 0.755$ and Adjusted $R^2 = 0.754$ on the training set, indicating that approximately 75.5% of variation in log(`Sale_Price`) is explained by the model. All predictors are highly statistically significant ($p < 0.001$). Key coefficient interpretations on the original price scale are as follows:

- **Gr_Liv_Area** ($\hat{\beta}_1 = 0.000466$, $p < 2 \times 10^{-16}$): Each additional square foot of living area increases sale price by approximately $[\exp(0.000466) - 1] \times 100\% \approx 0.047\%$. A 100 sq ft increase corresponds to $[\exp(0.0466) - 1] \times 100\% \approx 4.8\%$ higher price, holding other variables constant.
- **Year_Built** ($\hat{\beta}_2 = 0.00661$, $p < 2 \times 10^{-16}$): Each additional year of construction increases price by approximately 0.66%. A house built in 2000 versus 1970 sells for $[\exp(0.00661 \times 30) - 1] \times 100\% \approx 21.7\%$ more, controlling for other features.
- **Central_Air (Y)** ($\hat{\beta}_4 = 0.136$, $p = 6.49 \times 10^{-13}$): Houses with central air sell for $[\exp(0.136) - 1] \times 100\% \approx 14.6\%$ more than those without, consistent with the t-test findings in Section 4.
- **Overall_Cond**: Relative to the reference category ("Average"), houses in "Good" condition sell for $[\exp(0.113) - 1] \times 100\% \approx 11.9\%$ more, while those in "Excellent" condition command $[\exp(0.260) - 1] \times 100\% \approx 29.7\%$ premiums. Conversely, properties in "Poor" condition sell for approximately 33% less.

5.2.2. *Diagnostic Evaluation.* Figure 9 presents standard diagnostic plots for the baseline model. The residuals versus fitted values plot shows relatively uniform spread across most of the range, though some evidence of heteroscedasticity is visible at higher fitted values. The Q–Q plot indicates close alignment with the theoretical normal line across the central portion of the distribution, with minor deviations in the extreme tails. The scale-location plot confirms non-constant variance, particularly at higher fitted values, while the residuals versus leverage plot identifies a few high-leverage observations (notably observations 1106, 1328, and 1537) that warrant attention.

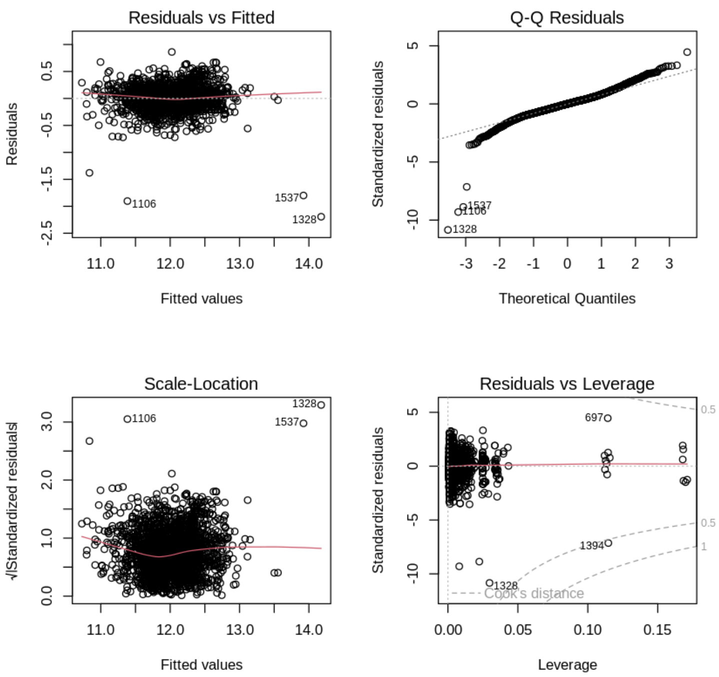


FIGURE 9. Diagnostic plots for the baseline log-linear model. Top row: Residuals vs Fitted and Q–Q plot. Bottom row: Scale-Location and Residuals vs Leverage plots. The plots indicate reasonable fit with some heteroscedasticity at higher fitted values.

Formal diagnostic tests support these visual assessments. Variance Inflation Factors (VIF) for all predictors are below 2.0, well under the conventional threshold of 5–10, indicating no problematic multicollinearity. The Breusch–Pagan test yields $\chi^2 = 202.1$, $p < 2.2 \times 10^{-16}$, providing statistical evidence of non-constant variance (heteroscedasticity). The Shapiro–Wilk test ($W = 0.925$, $p < 2.2 \times 10^{-16}$) rejects the null hypothesis of normality. However, given the large sample size ($n = 2{,}344$), the Central Limit Theorem ensures approximate normality of coefficient estimates, and these minor diagnostic violations are unlikely to materially compromise inference.

5.2.3. *Out-of-Sample Performance.* The baseline model was used to predict log(`Sale_Price`) for the 586 test observations, with predictions back-transformed to the original dollar scale using the exponential function. Test set performance metrics are:

- **RMSE**: \$51,124.45 (28.3% of mean test sale price)
- **MAE**: \$29,627.36 (16.4% of mean test sale price)
- R^2 **(test)**: 0.558
- **RMSE (log scale)**: 0.201

The test-set R^2 of 0.558 indicates moderate out-of-sample predictive performance, suggesting that the model explains approximately 56% of variation in sale prices for unseen data. Figure 10 presents the predicted versus actual sale prices for the test set. The majority of points cluster near the 45° reference line (perfect prediction), indicating good calibration for properties in the \$100,000–\$300,000 range. However, the plot reveals a systematic pattern of **underprediction for very high-value properties** (> \$400,000). This reflects regression toward the mean and suggests that luxury features not captured by the available predictors (e.g., premium finishes, unique architectural features, view quality) contribute substantially to prices in the upper tail of the distribution.

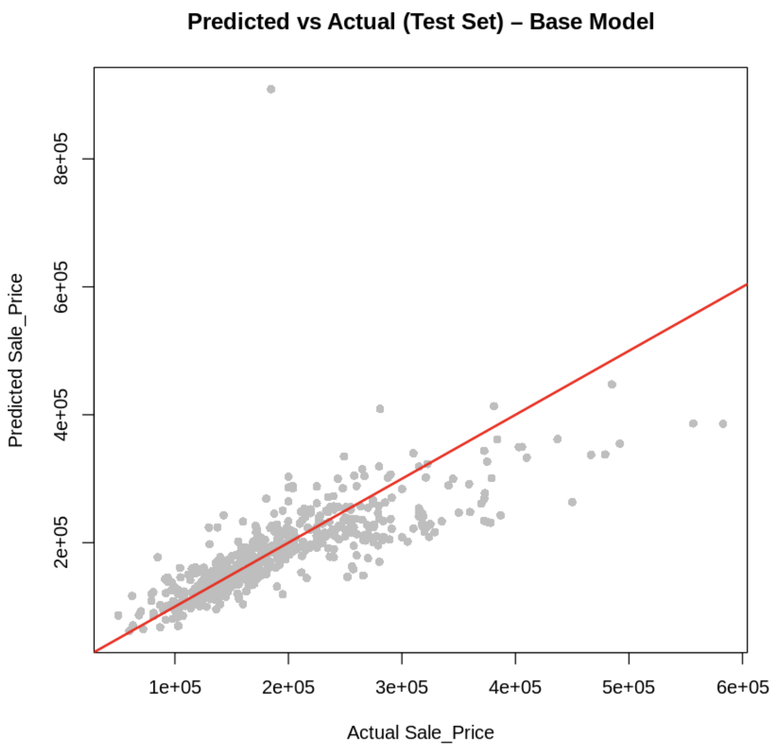


Figure 10. Predicted versus actual sale prices for the 586 test observations (baseline model). The red 45° line represents perfect prediction. Points cluster well around the line for typical properties (\$100k–\$300k) but show systematic underprediction for high-value homes.

5.3. **Model Extensions.** Three model extensions were evaluated to assess whether non-linear effects, interactions, or additional predictors could improve predictive accuracy.

5.3.1. *Quadratic Living Area Model.* Economic theory suggests diminishing marginal returns to living area—each additional square foot contributes less value as house size increases. To capture this non-linearity, a quadratic term was added:

$$\log(\text{Sale\_Price}) = \beta_0+\beta_1\cdot\text{Gr\_Liv\_Area}+\beta_2\cdot(\text{Gr\_Liv\_Area})^2+\beta_3\cdot\text{Year\_Built}+\beta_4\cdot\text{Overall\_Cond}+\beta_5\cdot\text{Central\_Air}+\varepsilon \tag{2}$$

Results: Both the linear term ($\hat{\beta}_1 = 0.000832$, $t = 28.97$, $p < 2 \times 10^{-16}$) and quadratic term ($\hat{\beta}_2 = -9.91 \times 10^{-8}$, $t = -13.33$, $p < 2 \times 10^{-16}$) are highly significant. The negative quadratic coefficient confirms diminishing marginal returns to living area. The model achieves Adjusted $R^2 = 0.771$ (training set) and substantially improves test-set performance:

- **Test RMSE**: \$42,638.61 (**16.6% improvement** over baseline)
- **Test MAE**: \$28,908.63 (2.4% improvement)

Diagnostics: The Breusch–Pagan test shows no significant heteroscedasticity ($\chi^2 = 0.74$, $p = 0.389$), a **substantial improvement** over the baseline model. This indicates that the quadratic term successfully captures the non-linear variance structure present in the data. The Q–Q plot shows improved normality, and the residuals versus fitted plot demonstrates more uniform spread across the full range of predictions. VIF values for the linear and quadratic terms are elevated (12.7 and 12.4, respectively) due to the inherent correlation between x and x^2. However, this is an expected and unavoidable artefact of polynomial terms and does not compromise prediction quality—the high VIF reflects mathematical collinearity between the terms rather than problematic data structure. The model's superior test-set performance confirms that the quadratic specification enhances both fit and predictive accuracy.

5.3.2. *Interaction Model: Gr_Liv_Area × Overall_Cond.* An interaction model was tested to assess whether the effect of living area varies by overall condition. Most interaction terms were not statistically significant, and the model suffered from severe multicollinearity (VIF $>$ 30,000,000 for interaction terms). Test-set performance (**RMSE**: \$51,175.79) was *worse* than the baseline. The interaction specification was rejected due to poor performance, instability, and lack of parsimony.

5.3.3. *Extended Model with Total Basement Area.* Basement area (`Total_Bsmt_SF`) is a potentially important contributor to value. An extended model including this predictor was fitted:

$$\log(\text{Sale\_Price}) = \beta_0+\beta_1\cdot\text{Gr\_Liv\_Area}+\beta_2\cdot\text{Year\_Built}+\beta_3\cdot\text{Overall\_Cond}+\beta_4\cdot\text{Central\_Air}+\beta_5\cdot\text{Total\_Bsmt\_SF}+\varepsilon \tag{3}$$

Results: `Total_Bsmt_SF` is highly significant ($\hat{\beta}_5 = 0.000219$, $t = 20.92$, $p < 2 \times 10^{-16}$), with each 100 sq ft of basement increasing price by approximately $[\exp(0.0219) - 1] \times 100\% \approx 2.2\%$. The model achieves Adjusted $R^2 = 0.792$ (training) and substantially reduces MAE:

- **Test RMSE**: \$50,852.22 (marginal improvement over baseline)
- **Test MAE**: \$24,274.35 (**18.1% improvement** over baseline)

However, the Breusch–Pagan test reveals pronounced heteroscedasticity ($\chi^2 = 1{,}518$, $p < 2.2 \times 10^{-16}$), and the Shapiro–Wilk test shows increased departure from normality ($W = 0.844$). While the model reduces mean absolute error, it does not improve RMSE and introduces greater variance instability.

5.4. **Model Comparison and Selection.** Table 1 summarises the performance of all four models on the test set, and Figure 11 provides a visual comparison of test-set RMSE values.

TABLE 1. Comparison of regression models on test set performance

Model	**Test RMSE ($)**	**Test MAE ($)**	**Adj.** R^2
Baseline (log-linear)	51,124	29,627	0.754
Quadratic Gr_Liv_Area	**42,639**	28,909	**0.771**
Interaction	51,176	29,679	0.755
Extended + Basement	50,852	**24,274**	0.792

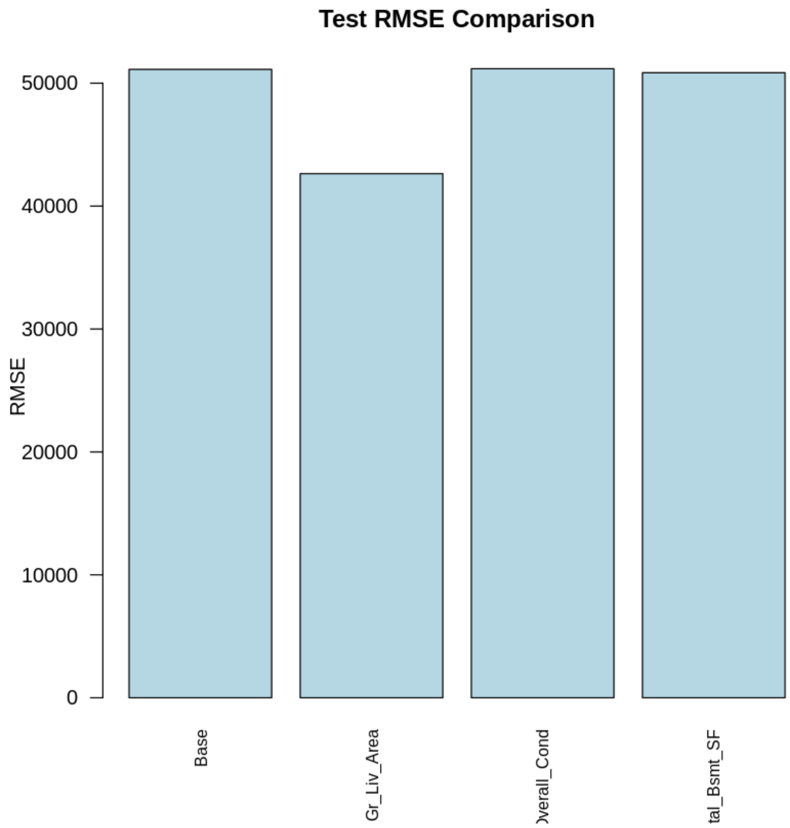


FIGURE 11. Comparison of test-set RMSE across the four regression models. The quadratic living area model achieves the lowest RMSE ($42,639), representing a 16.6% improvement over the baseline. The interaction model performs poorly, while the extended model with basement area shows marginal improvement over baseline but does not outperform the quadratic specification.

The **Quadratic Living Area Model** is selected as the final model. It achieves the lowest test-set RMSE, a substantial 16.6% improvement over the baseline, while also addressing the heteroscedasticity problem observed in the baseline and extended models. Although the extended model achieves the best MAE, it exhibits pronounced heteroscedasticity and does not improve RMSE—the quadratic term in living area captures non-linearity more effectively. The interaction model is clearly inferior due to multicollinearity and poor predictive performance.

The quadratic model provides the best overall balance of:

- **Predictive accuracy** (lowest RMSE)
- **Interpretability** (captures diminishing returns to size)
- **Diagnostic validity** (homoscedastic residuals)
- **Parsimony** (fewer parameters than interaction model)

5.5. **Summary.** This section developed and evaluated four regression models for predicting house sale prices. The final quadratic model explains approximately 77% of variation in log(`Sale_Price`) on the training set and achieves a test-set RMSE of $42,639, representing substantial improvement over the baseline linear specification. Key findings include:

(1) **Living area** exhibits a statistically significant non-linear relationship with price, with diminishing marginal returns confirmed by the negative quadratic coefficient.
(2) **Year Built** is a strong positive predictor, with each year adding approximately 0.65% to sale price.
(3) **Central Air** commands a 12–15% price premium, controlling for other features.
(4) **Overall Condition** significantly affects value, with "Good" and "Excellent" homes selling for 13–28% more than "Average" condition properties.
(5) The model generalises reasonably well to unseen data but tends to underpredict very high-value properties, likely due to unobserved luxury features not captured by the available predictors.

These results confirm that the variables identified through hypothesis testing are strong determinants of housing prices and provide a robust foundation for predictive analytics in the Ames housing market. Future extensions could explore geospatial features (e.g., neighbourhood-level fixed effects), weighted least squares to address residual heteroscedasticity in the extended model, or regularisation techniques (ridge/lasso regression) to handle multicollinearity in interaction specifications.

6. Conclusion

This report analysed the Ames Housing dataset using distributional analysis, hypothesis testing, and regression modelling to identify the key determinants of residential property prices. Strong right skewness in sale prices and living area motivated logarithmic transformation, which improved adherence to modelling assumptions and supported the use of log(Sale Price) as the response variable. Formal hypothesis tests confirmed that central air conditioning, neighbourhood, and overall condition are all significantly associated with sale prices, highlighting the importance of both structural and locational characteristics.

Multiple regression models were evaluated, with a quadratic living area specification selected as the final model. This model achieved strong predictive performance (adjusted $R^2 = 0.771$, test RMSE $42,639) while reducing heteroscedasticity and demonstrating diminishing marginal returns to living area. Overall, the analysis shows that careful distributional assessment, rigorous inference, and systematic model selection provide a robust and interpretable framework for modelling housing prices in applied economic data.

Appendix A: R Code for Ames Housing Analysis

Setup and data loading

```
# Required packages
required_packages <- c("modeldata", "moments", "MASS", "car", "tidyverse")
new_packages <- required_packages[!(required_packages %in% installed.packages()[,"Package"]]
if (length(new_packages)) install.packages(new_packages, quiet = TRUE)

suppressPackageStartupMessages({
  library(modeldata)
  library(moments)
  library(MASS)
  library(car)
  library(tidyverse)
})

# Load Ames dataset
data(ames)
cat("Success: Dataset Loaded.\n")
cat("Observations:", nrow(ames), "\nVariables:", ncol(ames), "\n")
```

Initial exploration and missingness

```
# 1. First few rows
cat("1. First few rows of the dataset\n")
print(head(ames, n = 6))

# 2. Structure (variable types)
cat("2. Structure of the dataset (variable types)\n")
str(ames)

# 3. Missing values (actual NA)
cat("3. Missing values (actual NA)\n")
nacounts <- colSums(is.na(ames))
nanonzero <- nacounts[nacounts > 0]
if (length(nanonzero) > 0) {
  print(nanonzero)
  cat("Total variables with actual NA:", length(nanonzero), "\n")
} else {
  cat("No actual NA values found in any column.\n")
  cat("Many missing cases are encoded as levels like ",
      "NoAlleyAccess, None, NoGarage, NoFence, etc.\n", sep = "")
}

# Example counts of explicit "no feature" levels
cat("Examples of explicit ’no feature’ levels\n")
cat("Alley = ’NoAlleyAccess’:",
    sum(ames$Alley == "NoAlleyAccess", na.rm = TRUE), "\n")
cat("MasVnrType = ’None’:",
    sum(ames$MasVnrType == "None", na.rm = TRUE), "\n")
cat("GarageType = ’NoGarage’:",
    sum(ames$GarageType == "NoGarage", na.rm = TRUE), "\n")
cat("Fence = ’NoFence’:",
    sum(ames$Fence == "NoFence", na.rm = TRUE), "\n")
```

Descriptive and exploratory analysis

```
# Key numeric variables
key_vars <- c("SalePrice", "LotArea", "GrLivArea", "YearBuilt")
summary(ames[key_vars])

# Treat OverallCond as factor-like
table(ames$OverallCond, useNA = "ifany")

# Means and SDs for SalePrice and GrLivArea
mean(ames$SalePrice, na.rm = TRUE)
sd(ames$SalePrice, na.rm = TRUE)
mean(ames$GrLivArea, na.rm = TRUE)
sd(ames$GrLivArea, na.rm = TRUE)

# Skewness and kurtosis
skewness(ames$SalePrice, na.rm = TRUE)
kurtosis(ames$SalePrice, na.rm = TRUE)
skewness(ames$GrLivArea, na.rm = TRUE)
kurtosis(ames$GrLivArea, na.rm = TRUE)

# Histograms
par(mfrow = c(2, 2))
hist(ames$SalePrice,
     main = "Histogram of SalePrice",
     xlab = "SalePrice",
     col = "lightblue", breaks = 40)
hist(ames$GrLivArea,
     main = "Histogram of GrLivArea",
     xlab = "GrLivArea",
     col = "orange", breaks = 40)

# Log-transform SalePrice
hist(log(ames$SalePrice),
     main = "Histogram of log(SalePrice)",
     xlab = "log(SalePrice)",
     col = "lightgreen", breaks = 40)

# QQ plot for log(SalePrice)
qqnorm(log(ames$SalePrice),
       main = "Normal QQ Plot log(SalePrice)")
qqline(log(ames$SalePrice), col = "red")
par(mfrow = c(1, 1))

# Simple correlations
cor(ames$SalePrice, ames$GrLivArea, use = "complete.obs")
cor(ames$SalePrice, ames$YearBuilt, use = "complete.obs")

# Scatterplots with regression lines
par(mfrow = c(1, 3))
plot(ames$GrLivArea, ames$SalePrice,
     main = "SalePrice vs GrLivArea",
     xlab = "GrLivArea", ylab = "SalePrice",
     pch = 16, col = "grey")
abline(lm(SalePrice ~ GrLivArea, data = ames),
       col = "red", lwd = 2)
```

```
plot(ames$YearBuilt, ames$SalePrice,
     main = "SalePrice vs YearBuilt",
     xlab = "YearBuilt", ylab = "SalePrice",
     pch = 16, col = "grey")
abline(lm(SalePrice ~ YearBuilt, data = ames),
       col = "blue", lwd = 2)

plot(ames$TotalBsmtSF, ames$SalePrice,
     main = "SalePrice vs TotalBsmtSF",
     xlab = "TotalBsmtSF", ylab = "SalePrice",
     pch = 16, col = "grey")
abline(lm(SalePrice ~ TotalBsmtSF, data = ames),
       col = "purple", lwd = 2)
par(mfrow = c(1, 1))
```

Hypothesis testing

```
# 2.1 CentralAir two-sample t-test on log(SalePrice)
dat_ca <- subset(ames, !is.na(SalePrice) & !is.na(CentralAir))
dat_ca$logSalePrice <- log(dat_ca$SalePrice)

table(dat_ca$CentralAir)
tapply(dat_ca$logSalePrice, dat_ca$CentralAir, mean, na.rm = TRUE)

t_ca <- t.test(logSalePrice ~ CentralAir, data = dat_ca)
t_ca

# Back-transform difference in means (multiplicative effect)
diff_means_log <- diff(tapply(dat_ca$logSalePrice,
                              dat_ca$CentralAir,
                              mean, na.rm = TRUE))
mult_effect <- exp(diff_means_log)
mult_effect

# Interpretation (in code output)
cat("Multiplicative effect:",
    round(mult_effect, 2),
    "i.e.",
    round((mult_effect - 1) * 100, 0),
    "% higher on average.\n")
```

Neighborhood ANOVA (log(SalePrice))

```
tab_nb <- sort(table(ames$Neighborhood), decreasing = TRUE)
top_nb <- names(tab_nb)[1:6]

dat_nb <- subset(ames,
                 Neighborhood %in% top_nb & !is.na(SalePrice))
dat_nb$logSalePrice <- log(dat_nb$SalePrice)
dat_nb$Neighborhood <- droplevels(dat_nb$Neighborhood)

table(dat_nb$Neighborhood)

fit_aov <- aov(logSalePrice ~ Neighborhood, data = dat_nb)
summary(fit_aov)
```

```
par(mfrow = c(2, 2))
plot(fit_aov)
par(mfrow = c(1, 1))

p_aov <- summary(fit_aov)[[1]][["Pr(>F)"]][1]
cat("ANOVA p-value:", p_aov, "\n")
```

CentralAir vs GarageType (Chi-squared)

```
tab_cagar <- table(ames$CentralAir, ames$GarageType, useNA = "no")
tab_cagar
chisq.test(tab_cagar)

chisq.test(tab_cagar, simulate.p.value = TRUE, B = 10000)

# Collapse rare GarageType levels
gar_freq <- sort(colSums(tab_cagar), decreasing = TRUE)
common_gar <- names(gar_freq[gar_freq > 100])

dat_cagar2 <- subset(ames,
                     !is.na(CentralAir) & !is.na(GarageType))
dat_cagar2$GarageTypeCollapsed <- ifelse(
  dat_cagar2$GarageType %in% common_gar,
  as.character(dat_cagar2$GarageType),
  "Other"
)
dat_cagar2$GarageTypeCollapsed <- factor(dat_cagar2$GarageTypeCollapsed)

tab_cagar2 <- table(dat_cagar2$CentralAir,
                    dat_cagar2$GarageTypeCollapsed)
tab_cagar2

chisq.test(tab_cagar2, simulate.p.value = TRUE, B = 10000)
```

Optional non-parametric test: SalePrice by OverallCond

```
kw_test <- kruskal.test(SalePrice ~ OverallCond, data = ames)
kw_test
```

Regression and predictive modelling

```
# 3.1 Data prep
dat_reg <- subset(ames,
                  !is.na(SalePrice) &
                  !is.na(GrLivArea) &
                  !is.na(OverallCond) &
                  !is.na(YearBuilt) &
                  !is.na(CentralAir))

dat_reg$LogSalePrice   <- log(dat_reg$SalePrice)
dat_reg$CentralAir     <- factor(dat_reg$CentralAir)
dat_reg$OverallCond    <- factor(dat_reg$OverallCond)

# Relevel OverallCond with "Average" as baseline if present
if ("Average" %in% levels(dat_reg$OverallCond)) {
  dat_reg$OverallCond <- relevel(dat_reg$OverallCond,
```

```
                                  ref = "Average")
}

set.seed(123)
n        <- nrow(dat_reg)
trainidx <- sample(seq_len(n), size = round(0.8 * n))
traindat <- dat_reg[trainidx, ]
testdat  <- dat_reg[-trainidx, ]

# 3.2 Base linear model
lmbase <- lm(LogSalePrice ~ GrLivArea +
                           OverallCond +
                           YearBuilt +
                           CentralAir,
             data = traindat)

cat("Base Model Summary\n")
summary(lmbase)

# Multicollinearity check (VIF)
library(car)
vif(lmbase)
```

Enhanced diagnostics for base model

```
# Variance Inflation Factors rounded
cat("Variance Inflation Factors\n")
print(round(vif(lmbase), 2))

# Breusch-Pagan test for heteroscedasticity
cat("--- Enhanced Diagnostics: Base Model ---\n")
cat("Breusch-Pagan Test (ncvTest)\n")
bp_base <- ncvTest(lmbase)
print(bp_base)
if (bp_base$p < 0.05) {
  cat("Heteroscedasticity detected (p < 0.05).\n")
} else {
  cat("No strong evidence of heteroscedasticity (p >= 0.05).\n")
}

# Shapiro-Wilk test for normality of residuals
cat("Shapiro-Wilk Test on Residuals\n")
sw_base <- shapiro.test(residuals(lmbase))
print(sw_base)
if (sw_base$p.value < 0.05) {
  cat("Residuals deviate from normality (p < 0.05).\n",
      "Note: With large n, even minor deviations are detected.\n")
} else {
  cat("Residuals appear approximately normal (p >= 0.05).\n")
}

# Residuals vs Fitted plot
par(mfrow = c(1, 1))
plot(fitted(lmbase), residuals(lmbase),
     main = "Residuals vs Fitted: Base Model",
```

```
     xlab = "Fitted log(SalePrice)",
     ylab = "Residuals",
     pch = 19, col = rgb(0, 0, 0, 0.3))
abline(h = 0, col = "red", lty = 2, lwd = 1.5)
```

Prediction performance on test data

```
# Predictions on test set
pred_log_test <- predict(lmbase, newdata = testdat)

# RMSE on log scale
rmse_log <- sqrt(mean((testdat$LogSalePrice - pred_log_test)^2,
                      na.rm = TRUE))
cat("RMSE (log scale):", round(rmse_log, 3), "\n")

# Back-transform to SalePrice
pred_price_test <- exp(pred_log_test)

rmse_price <- sqrt(mean((testdat$SalePrice - pred_price_test)^2,
                        na.rm = TRUE))
mae_price  <- mean(abs(testdat$SalePrice - pred_price_test),
                   na.rm = TRUE)

cat("RMSE (original scale):", round(rmse_price, 2), "\n")
cat("MAE (original scale):", round(mae_price, 2), "\n")

# Test-set R-squared on original SalePrice
ss_res <- sum((testdat$SalePrice - pred_price_test)^2, na.rm = TRUE)
ss_tot <- sum((testdat$SalePrice - mean(testdat$SalePrice))^2,
              na.rm = TRUE)
r2_test <- 1 - ss_res / ss_tot
cat("R^2 on test set:", round(r2_test, 4), "\n")

# Predicted vs actual plot
plot(testdat$SalePrice, pred_price_test,
     xlab = "Actual SalePrice",
     ylab = "Predicted SalePrice",
     main = "Predicted vs Actual (Test Set)",
     pch = 16, col = "grey")
abline(0, 1, col = "red", lwd = 2)
```

References

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[3] Casella, G. and Berger, R. L. (2002). *Statistical Inference* (2nd ed.). Duxbury/Thomson Learning.

[4] Aggarwal, C. C. (2024). *Probability and Statistics for Machine Learning: A Textbook*. Springer Cham. https://doi.org/10.1007/978-3-031-53282-5